

STAT 408 – Final Report

Students in STAT408 are required to find a medium sized data set (i.e. more than ~1000 observations but small enough that your computer can handle computation (i.e. don't pick a data set with a billion observations)).

Student will write a paper summarizing presenting their results. To facilitate timeliness of grading, the paper must be a maximum length of 5 pages, not including any tables or figures.

You should also make your code available via Github or another publicly available repository.

The paper should include the following sections:

1. Introduction
 - a. Provide a clear overview of your data set
 - b. Explain why what you are doing is interesting (This section should essentially be all words.)
2. Data Exploration and Preparation
 - a. Conducts comprehensive data exploration.
 - b. Describes the dataset used, including its source and characteristics.
 - c. Discusses any preprocessing steps undertaken, such as data cleaning or feature engineering.
 - d. Presents summary statistics and visualizations to aid understanding.
3. Model Building
 - a. Clearly explains the process of model building.
 - b. Describes the selection of independent variables and the rationale behind their inclusion.
 - c. Discusses the choice of model selection criteria (e.g., R-squared, AIC, BIC).
 - d. Provides insights into any model diagnostics performed (e.g., checking for multicollinearity, heteroscedasticity).
4. Interpretation of Results
 - a. Presents the results of the linear regression analysis clearly and concisely.
 - b. Interprets the coefficients of the regression model, including their significance and direction.
 - c. Discusses the overall fit of the model and its predictive power.
 - d. Provides actionable insights based on the regression results.
5. Evaluation and Discussion
 - a. Critically evaluates the strengths and weaknesses of the linear regression analysis.
 - b. Discusses any limitations of the study, such as data constraints or model assumptions.
 - c. Considers alternative approaches or models that could improve the analysis.
 - d. Reflects on the implications of the findings and their relevance to the broader field.
 - e. Describe what you would have done if you had more time (i.e. future work).